COMPUTER CLASSES

Lecture 01



All ABOUT PDF

TRIPLE S

-By Er Arshad Ishaq Chouhan

➢ **TABLE OF CONTENTS**

1. INTRODUCTION
2. ADVANTAGES AND DISADVANTAGES OF PDF
3. TYPES OF PDF
4. COMMON PDF SOFTWARE



1. INTRODUCTION

- **Portable Document Format (PDF)** is the official full form of PDF.
- Developed by **Adobe Systems** (led by Dr. John Warnock under "The Camelot Project") in **1992**; launched in 1993, containing text, images, and graphics to be displayed consistently across different application software, hardware, and operating systems.
- Transitioned from proprietary to a fully **Open Standard** in 2008. It is officially maintained as **ISO 32000**.
- In simple terms, PDF is a secure file format used to present and share documents independently of the device or software being used.
- A PDF file preserves all the elements of a printed document in a digital format that can be viewed, navigated, printed, or shared with others.
- PDF is directly based on the PostScript page description language (also developed by Adobe). PostScript tells the printer exactly where to drop ink on a page; PDF simply takes that language and optimizes it for onscreen viewing.
- **MIME Type:** If a question asks about the internet media type for a PDF, it is application/pdf.

➢ **SOME COMMON USES OF PDF INCLUDE:**

- ✓ Searching for specific information within a PDF file.
- ✓ Copying data from a PDF document.
- ✓ Modifying pages by adding, deleting, or rearranging them.
- ✓ Making minor text corrections, such as fixing spelling or formatting errors.
- ✓ Merging multiple files of different formats into a single PDF document.

➢ ARCHITECTURE & FILE STRUCTURE

A PDF operates on an internal object architecture called **COS** (*Carousel Object Structure*).A standard PDF file structure is divided sequentially into four main parts:

i. **Header:** The very first line of the file. It specifies the version of the PDF standard being used (e.g., %PDF-1.7).

ii. **Body:** Contains all the visible components of the document. This includes text streams, vector shapes, raster images, and references to fonts.

iii. **Cross-Reference (XRef) Table:** A critical internal index that lists the exact byte offsets of every object inside the file. This allows PDF reader software to jump directly to any page or image instantly without reading the entire file sequentially.

iv. **Trailer:** The final lines of the file. It tells the reader application how to find the XRef table and securely read the document from the end.

❑ Note: PDF reader application parses a PDF document **from the bottom up**, reading the Trailer first to unlock the XRef table.

➢ CONTENT TYPES INSIDE THE CONTAINER

- **Text Streams:** Stores character data alongside precise geometric layout coordinates to lock formatting in place.
- **Vector Graphics:** Paths and Bézier curves used for shapes and logos; scales infinitely without pixelation.
- **Raster Images:** Pixel-based photos using built-in compression filters:
  - ✓ FlateDecode: Lossless compression (similar to .zip).
  - ✓ DCTDecode: Lossy compression (based on .jpeg).

➢ THE FOUR FUNDAMENTAL COS OBJECTS

- **Dictionaries (<< >>):** Key-value pairs enclosed in double angled brackets that define object properties (such as page dimensions or rotation rules).
- **Streams:** Large blocks of binary data (like raw images or compressed page drawing instructions) paired with a dictionary explaining its compression filter.
- **Arrays ([ ]):** Ordered collections of objects enclosed in square brackets, frequently used to define coordinate bounding boxes (e.g., standard page layouts).
- **Indirect Objects (obj):** Elements assigned a unique ID number so they can be referenced and reused across multiple pages (like a corporate logo), preventing file size inflation.

➢ TYPOGRAPHY & FONT MANAGEMENT

To ensure text looks exactly the same on every computer, PDFs handle typography using three specific protocols:

- **Fully Embedded Fonts:** The file packages the *entirety* of the font file (every letter, number, and glyph) inside its database. This ensures perfect rendering and full downstream editing capabilities but makes the file size significantly larger.
- **Font Subsetting:** The file packages only the specific characters used in that precise document. If a heading only uses the letters X, Y, and Z, only those three glyphs are saved. This drastically reduces file size while preserving exact visuals, though it restricts later editing.
- **Font Substitution (The Fallback System):** If a font is neither embedded nor installed on the user's computer, the reader app falls back on default fonts (**AdobeSerifMM** or **AdobeSansMM**), stretching or compressing them mathematically to perfectly preserve the original line breaks and page geometry.
- **The "Standard 14" Fonts:** Early ISO specifications established 14 foundational fonts that every compliant PDF reader must natively recognize without needing them embedded:
  - ✓ Times Family: Times-Roman, Times-Bold, Times-Italic, Times-BoldItalic
  - ✓ Helvetica Family: Helvetica, Helvetica-Bold, Helvetica-Oblique, Helvetica-BoldOblique
  - ✓ Courier Family: Courier, Courier-Bold, Courier-Oblique, Courier-BoldOblique
  - ✓ Symbolic: Symbol, ZapfDingbats

➢ PURPOSE OF CREATION

- The main goals behind creating PDF were:
  - ✓ To solve compatibility issues between different computer systems.
  - ✓ To maintain consistent document formatting.
  - ✓ To support reliable digital document sharing and printing.

➢ KEY FEATURES/ADVANTAGES OF PDF

- ✓ **Fixed Layout** – Formatting does not change.
- ✓ **Platform Independence:** Guarantees uniform rendering across Windows, macOS, Linux, Android, and iOS without font or layout displacement.
- ✓ **Read-Only Integrity:** Designed natively for reading and distribution rather than active content modification.
- ✓ **Dual-Layer Security:**
  - ✓ User Password (Open Password): Restricts unauthorized viewing/opening of the file.
  - ✓ Owner Password (Permissions Password): Restricts explicit actions such as printing, content copying, or editing.
- ✓ **Cryptographic Authentication:** Supports digital signatures to verify document source integrity and detect unauthorized tampering.
- ✓ **Interactive Framework:** Capable of embedding form fields, hyperlinks, buttons, and multimedia assets.
- ✓ **Compression** – Reduces file size.
- ✓ **Searchable Text** – Easy content search within document.
- ✓ **The "Print to PDF" Concept:** This feature functions as a **virtual printer**. Instead of sending data to physical paper via ink, the operating system routes the print stream into a digital file container, locking the layout exactly as it would look on paper.

DISAVANTAGES OF PDF

- ✓ Difficult to edit directly
- ✓ Some advanced tools are paid
- ✓ Large multimedia PDFs may open slowly
- ✓ Editing scanned PDFs requires OCR

4. TYPES OF PDF

- **True or Standard PDF**
  - ✓ Regular digital document. e.g. PDF created by Ms-Word, Ms-Excel etc
  - ✓ Easily modifiable, searchable.
- **Scanned PDF or Image only PDF**
  - ✓ Image-based document.
  - ✓ The pdfs which are created with the help of scanners.
  - ✓ Not searchable, editing is difficult.

- **Scanned PDF or Image only PDF**
  - ✓ Image-based document.
  - ✓ The pdfs which are created with the help of scanners.
  - ✓ Not searchable, editing is difficult.

- ✓ **Editable PDF** – Fillable forms or modifiable content.
- ✓ **Secured PDF** – Password-protected.
- ✓ **PDF/A** – For long-term archiving.
- ✓ **PDF/X** – Used in printing industry.
- ✓ **PDF/E** – Used in engineering documents.
- ✓ **PDF/UA** – PDF for **Universal Accessibility**

✓ COMMON PDF SOFTWARES

- ✓ Adobe Acrobat, pdfFiller by airSlate, Conga Composer, Foxit PDF Editor, Wondershare PDFelement,
- ✓ Inkscape, MadCap Flare, Microsoft Word, Sumatra PDFWeb Browser,
- ✓ Nitro ReaderSLim PDF, Expert PDF reader, Javelin PDF Reader, PDF-X-Change Editor, Google Drive

❑ NOTE:

- ✓ File Extension: .pdf
- ✓ Developed By: Adobe
- ✓ First Release: 15 June, 1993 (1.0 Version).
- ✓ ISO Standard: 2008 (Proprietary Software).
- ✓ Standard Number: ISO 32000
- ✓ Platform: Cross-platform
- ✓ Supports: Text, Graphics, Images, Forms, Multimedia
- ✓ The **latest official version of the PDF standard** is **PDF 2.0**, also known as **ISO 32000-2:2020**. This specification was published by the **International Organization for Standardization (ISO)** in 2020 and represents the most up-to-date formal definition of the Portable Document Format.

ARSHAD ISHAQ CHOUHAN

1. Which company originally developed the Portable Document Format (PDF) in the early 1990s?

A) Microsoft
B) Adobe
C) Apple
D) IBM

2. In which year was the PDF format first introduced to the public?

A) 1991
B) 1993
C) 1995
D) 1998

3. What was the internal project codename at Adobe that eventually led to the creation of the PDF?

A) Project Acrobat
B) Camelot
C) PostScript II
D) Velvet

4. Which international organization currently maintains the PDF format as an open standard?

A) W3C
B) IEEE
C) ISO
D) IETF

5. What is the standard file extension used for Portable Document Format files?

A) .pdx
B) .pdf
C) .pfd
D) .fdf

6. Which core component of the PDF file structure contains references to all the objects within the file, allowing random access?

A) Header
B) Body
C) Cross-reference (Xref) table
D) Trailer

7. Which specific PDF standard is strictly designed for long-term archiving and preservation of electronic documents?

A) PDF/X
B) PDF/A
C) PDF/E
D) PDF/UA

8. What is the primary purpose of the PDF/UA standard

A) Universal Archiving
B) User Authentication
C) Universal Accessibility
D) Engineering Data Exchange

9. Which programming language, also developed by Adobe, forms the foundational baseline for how PDF describes page layouts?

A) JavaScript
B) ActionScript
C) PostScript
D) HTML

10. What does the "Trailer" section of a PDF file primarily do?

A) It stores the compressed images of the document.
B) It tells the reader application how to start reading the file.
C) It contains the digital signature certificates.
D) It holds the metadata about the author.

11. Consider the following statements regarding the PDF format:

1. A PDF file is completely independent of the hardware, operating system, and application software used to create it.
2. PDF files cannot contain interactive elements like hyperlinks, buttons, or form fields.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

2. Consider the following statements regarding font embedding in PDFs:

1. Embedding a font ensures that the document looks the same on any device, even if the target device doesn't have that font installed.
2. "Subset embedding" includes only the specific characters used in the document, which reduces the overall file size.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

13. Consider the following statements regarding PDF security:

1. User Passwords (Open Passwords) restrict unauthorized users from opening and viewing the PDF file.
2. Owner Passwords (Permissions Passwords) are used to restrict actions like printing, copying text, or modifying the document.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

14. Consider the following statements regarding the PDF/X standard:

1. PDF/X is optimized specifically for blind digital data exchange in the printing and graphic arts industries.
2. PDF/X files strictly forbid the usage of color management profiles like CMYK.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

15. Consider the following statements regarding PDF structures:

1. A Linearized PDF (often called "Fast Web View") allows the first page of a PDF to be displayed in a web browser before the entire file finishes downloading.
2. Non-linearized PDFs must be fully downloaded by a browser before any page can be viewed.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

16. Consider the following statements regarding the evolution of PDF:

1. Adobe released the PDF specification free of charge in 1993, but it remained a proprietary format until it was published as an ISO standard in 2008.
2. ISO 32000-1 is the official designation for the PDF 1.7 specification.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

17. Consider the following statements regarding "Tagged PDF":

1. Tagged PDFs contain logical structure metadata that defines the reading order of text blocks, headings, and images.
2. Screen readers for the visually impaired rely heavily on Tagged PDFs to navigate the document correctly.

Which of the statements given above is/are correct?

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Which of the statements given above is/are correct?

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B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

18. Consider the following statements regarding PDF/A restrictions:

1. To ensure long-term preservation, PDF/A documents strictly forbid the use of audio, video, and external references.
2. JavaScript code and executable files are permitted in a PDF/A file as long as they are encrypted.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only C)
C) Both 1 and 2
D) Neither 1 nor 2

19. Consider the following statements regarding PDF compression:

1. PDF files can use both lossless compression (like Flate/ZIP) for text and lossy compression (like JPEG) for photographic images.
2. Compressing a PDF will always convert vector graphics into raster images, lowering the quality of the document.

Which of the statements given above is/are correct?

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B) 2 only
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C) Both 1 and 2
D) Neither 1 nor 2

20. Consider the following statements regarding PDF/E:

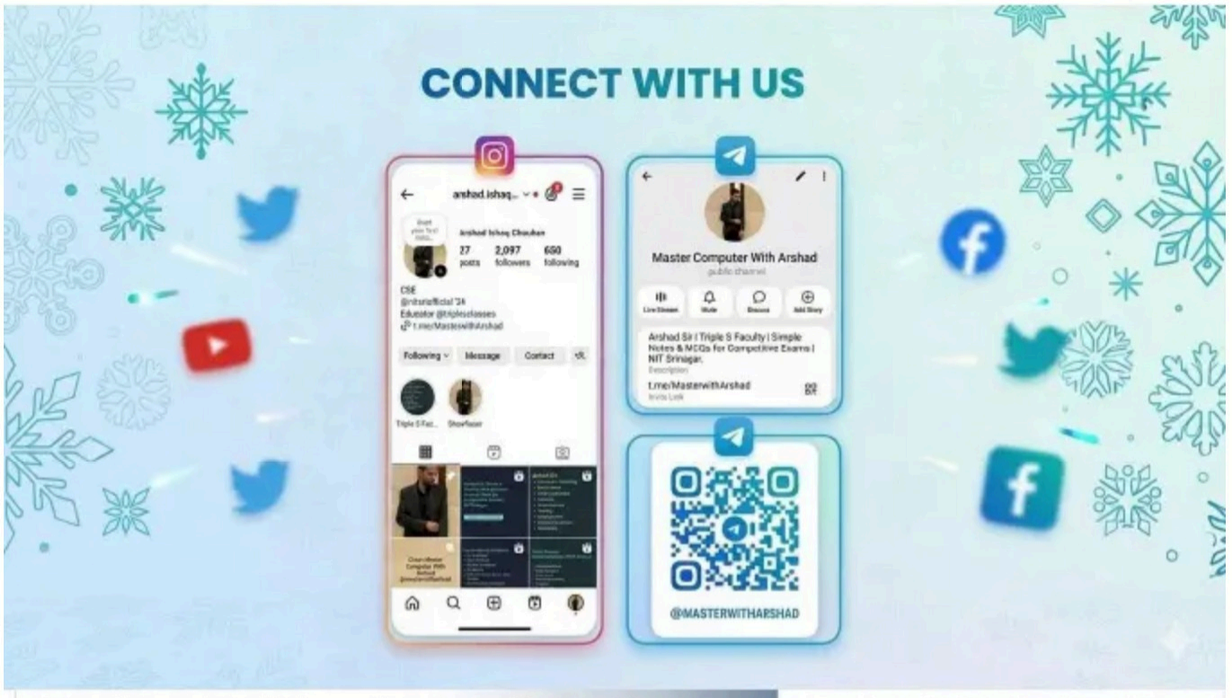
1. PDF/E is a standard specifically tailored for engineering, construction, and architectural workflows.
2. PDF/E supports the exchange of 3D drawings and interactive engineering multimedia.

Which of the statements given above is/are correct?

A) 1 only
B) 2 only
C) Both 1 and 2
D) Neither 1 nor 2

ANSWER KEY

Q.No	1	2	3	4	5	6	7	8	9	10
ANS	B	B	B	C	B	C	B	C	C	B
Q.No	11	12	13	14	15	16	17	18	19	20
ANS	A	C	C	A	C	C	C	A	A	C



Email: arshadishaqchouhan@gmail.com

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